SOCIOL 723: Social Statistics II

Week 14, Synthesis

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Today

- A synthesis: what the semester was actually about.
- A decision guide you can use on your own data.
- The final project: expectations, common failure modes, and the workshop.

Logistics

No lab Thursday November 26 (Thanksgiving recess).
Final project due December 14.

Synthesis

Three Questions, in Order

1. **What am I trying to estimate?** A unit-specific quantity and a target population. ATE, ATT, LATE, the effect at a cutoff, $\tau(x)$. Weeks 5 and 13.
2. **What would have to be true for my estimate to recover it?** Ignorability, exclusion, continuity, parallel trends. Weeks 5–11.
3. **How do I estimate it, and how uncertain am I?** Least squares, maximum likelihood, weighting, orthogonal scores, cross-fitting. Weeks 1–4, 6, 12–13.

The ordering is the lesson

Most quantitative papers start at 3 and never articulate 1 or 2. Most methodological disputes are really disagreements at 1 or 2 being conducted as if they were about 3.

One Idea, Recurring

FWL (Wk 1)
residualize D on W

Weighting / AIPW (Wk
residualize, then reweight

Flexible nuisances (Wk 13)
the same idea, fewer
functional-form claims

- **Partialling out** is the through-line. Remove the part of the treatment explained by the controls; use what is left.
- Its virtue is *insensitivity*: small errors in how you model the controls do not contaminate the parameter you care about.
- Everything else, matching, weighting, IV, RD, DID, synthetic control , is a different answer to the question *where does the counterfactual come from?*

Every Design, in One Table (cont.)

Design	Key assumption	Estiman
Regression / matching / IPW	ignorability given X	ATE, A
AIPW / DML	ignorability + rates	ATE, $\tau($
IV / 2SLS	relevance, independence, exclusion, monotonicity	LATE (
RD (sharp)	continuity at c	effect at
RD (fuzzy)	+ first-stage jump, monotonicity, exclusion	complie
DID (2×2)	parallel trends, no anticipation	ATT
DID, staggered	cohort-wise parallel trends vs. not-yet-treated, no anticipation	ATT(g,
Synthetic control	factor model, long pre-period	ATT fo

Every Design, in One Table (cont.)

- No row is safer than the others in the abstract. Credibility is a property of the application, not of the method.
- Notice how often the estimand is chosen *by the design* rather than by the researcher. Say so when it is.

Inference, in One Table (cont.)

Situation	Use	Week
Default	HC2 or HC3 robust	2
Treatment assigned in groups	cluster on the assignment level	2
Fewer than $\sim$40 clusters	CR2 + Satterthwaite, or wild cluster bootstrap	2
Nonlinear quantity of interest	delta method, simulation, or bootstrap	2, 4
Nuisance function estimated	bootstrap the whole pipeline, or an influence function	6
Weak instrument	Anderson–Rubin	7
One treated unit	permutation / placebo inference	11
ML nuisances	orthogonal score + cross-fitting	13

The recurring error is a standard error that ignores a step in the procedure, estimated weights, a selected model, a matched design, a chosen bandwidth. When in doubt,

Inference, in One Table (cont.)

bootstrap the whole pipeline, resampling in a way that *reproduces the sampling or assignment structure* (clusters as clusters, panels as panels). But the bootstrap is not universal: it is *invalid* for nearest-neighbour matching (Week 6), and unreliable more generally for non-smooth, boundary, post-selection, and weakly identified procedures.

A Decision Guide

1. Write the theoretical estimand in one sentence: unit-specific quantity, target population.
2. Draw the DAG. Mark what is unobserved.
3. Ask: *is there any set I can measure that closes the back doors?*
   - Yes → selection on observables. Check overlap, use a doubly robust estimator, report a sensitivity analysis (Week 6).
   - No → look for design-based variation: an instrument (7), a threshold (9), a policy change over time (10), a single treated unit with a long pre-period (11).
4. Check whether the design's estimand is the one you wrote in step 1. If not, say so explicitly rather than quietly redefining the question.
5. Choose inference to match the design.
6. If W is high-dimensional or the functional form is a guess, use an orthogonal score with cross-fitting (13).
7. Report what would change your mind.

Things That Are Never a Solution

- Adding more controls without a DAG. Bad controls exist, and "kitchen sink" amplifies bias as often as it reduces it (Week 5).
- Robust standard errors for a problem of bias (Week 2).
- A high R^2, or a high AUC, as evidence of causality (Weeks 1, 12).
- A non-significant specification test, balance t-tests, Sargan, pre-trend tests, as evidence that an assumption holds. Low power is not confirmation (Weeks 6, 7, 10).
- Machine learning for an identification problem (Week 13).
- Reporting whichever specification worked. Pre-commit, then show robustness.

Each of these is a real move made in published sociology, and you now have the vocabulary to say precisely why it fails.

Final Project

What I Am Asking For

A replication *and extension* of a published study, roughly 2,000–3,000 words plus reproducible `R` code. Due **December 14**.

1. **State the paper's estimand** in the Lundberg–Johnson–Stewart form. Often the paper does not, and stating it is already a contribution.
2. **Reproduce the headline result.** Report honestly if you cannot; that is a finding, not a failure.
3. **Identify the identifying assumption** and say where it is defended, or where it is merely asserted.
4. **Extend.** A sensitivity analysis; a modern estimator where the paper used TWFE; robust or clustered inference where it was missing; heterogeneity done honestly; a DML version relaxing a functional form.
5. **Say what would change your mind.**

Common Failure Modes

- **Choosing a paper without replication data.** Confirm availability *before* committing; this was the November 2 deadline.
- **Extending in a direction the data cannot support**: a causal forest on 300 observations, a DID with two periods and no pre-trend, an IV with a first-stage F of 4.
- **Reporting a battery of estimators with no argument.** Six numbers without a story is worse than one number with a defence.
- **Burying the estimand.** If I cannot tell from your first paragraph what quantity you are estimating and for whom, start again.
- **Unreproducible code.** Restart `R`, render from scratch, set seeds.

Scope down rather than up. A careful, well-defended small extension beats an ambitious one you cannot support.

Workshop Format

For the rest of today, in turn:

- **Two minutes**: the paper, the estimand in one sentence, and the identifying assumption.
- **Two minutes**: your planned extension and why the data can support it.
- **Five minutes**: questions from the room. Everyone should come with one question for someone else.

What to bring

One slide, or nothing at all. What matters is that you can say the estimand out loud without reading it.

Office hours Wednesdays 2–4PM, or by appointment, through December 14.

Where to Go Next (cont.)

- **Depth on design**: Angrist and Pischke, *Mostly Harmless Econometrics*; Cunningham, *Causal Inference: The Mixtape*; Huntington-Klein, *The Effect*.
- **Depth on potential outcomes**: Imbens and Rubin, *Causal Inference for Statistics, Social, and Biomedical Sciences*; Morgan and Winship.
- **Depth on graphs**: Pearl, *Causality*; Hernán and Robins, *Causal Inference: What If*.
- **Depth on causal ML**: Chernozhukov et al., *Applied Causal Inference Powered by ML and AI*.
- **In this department**: measurement and latent variables, computational text analysis, network methods, demographic methods, multilevel models, each of which assumes what you now know.

Where to Go Next (cont.)

The one thing to keep

Say what you are estimating, say what would have to be true, and say what would change your mind. Everything else is technique.